

Bus Arbitration.

Bus Arbitration refers to the process by which the current bus master accesses and then leaves the control of the bus and passes it to another bus requesting processor unit. The controller that has access to a bus at an instance is known as a Bus master.

→ A conflict may arise if the number of DMA controllers or other controllers or processors try to access the common bus at the same time, but access can be given to only one of those, only one processor or controller can be bus master at the same point in time. To resolve these conflicts, the bus arbitration procedure is implemented to coordinate the activities of all devices requesting memory transfers. The selection of the bus master must take into account the needs of various devices by establishing a priority system for gaining access to the bus. The bus arbiter decides who would become the current bus master.

→ Centralized bus arbitration: A single bus arbiter performs the required arbitration.

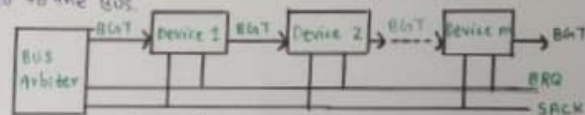
→ Distributed bus arbitration: - All devices participating in the selection of the next bus master.

Methods/Types of Centralized Bus Arbitration

There are three bus arbitration methods:-

1. Daisy chaining method.

→ It is a simple and cheaper method where all the bus masters use the same line for making bus requests. The bus grant signal serially propagates through each master until it encounters the first one that is requesting access to the bus. This master blocks the propagation of the bus grant signal, therefore any other requesting module will not receive the grant signal and hence cannot access the bus. During any bus cycle, the bus master may be any device - the processor or any DMA controller unit, connected to the bus.



Daisy chained Bus arbitration

Advantages:

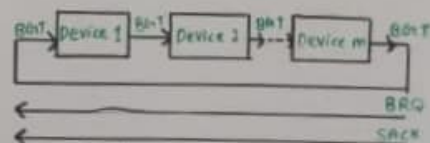
- Simplicity and scalability
- The user can add more devices anywhere along the chain, up to a certain maximum value.

Disadvantages:

- The value of priority assigned to a device depends on the position of the master bus
- Propagation delay arises in this method

2. Polling or Rotating Priority method.

→ In this, the controller is used to generate the address for the master (unique priority), the number of address lines required depends on the number of masters connected in the system. The controller generates a sequence of master addresses. When the requesting master recognizes its address, it activates the busy line and begins to use the bus.



Rotating Priority Bus arbitration

Advantages:

- This method does not favor any particular device and processor
- The method is also quite simple.

Disadvantages:

- Adding bus masters is difficult as it increases the number of address lines of the circuit.
- If one device fails then the entire system will not stop working.

3. Fixed Priority or Independent Request method.

→ In this, each master has a separate pair of bus request and bus grant lines and each pair has a priority assigned to it. The built-in priority decoder within the controller selects the highest priority request and asserts the corresponding bus grant signal.

Advantages:

- This method generates a fast response

Disadvantages:

- Hardware cost is high as a large no. of control line is required.

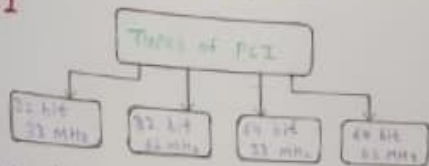


Fixed Priority bus arbitration method.

PCI Stands for Peripheral Component Interconnect

Peripheral Component Interconnect (PCI) is for Peripheral Component Interconnect. A Peripheral Component Interconnect Bus (PCI bus) connects the CPU and expansion boards such as modem cards, network cards and sound cards. These expansion boards are normally plugged into expansion slots on the motherboard. The PCI local bus is the general standard for a PC expansion bus, having replaced the Video Electronics Standards Association (VESA) local bus and the Industry Standard Architecture (ISA) bus. PCI has largely been replaced by USB.

Types of PCI



- PCI 32 bits have a transport speed of 33 MHz and work at 132 MBPS.
PCI 64 bits have a transport speed of 33 MHz and work at 264 MBPS.
PCI 32 bits have a transport speed of 66 MHz and work at 512 MBPS.
PCI 64 bits have a transport speed of 66 MHz and work at 1.0 GBPS.

Functions of PCI:

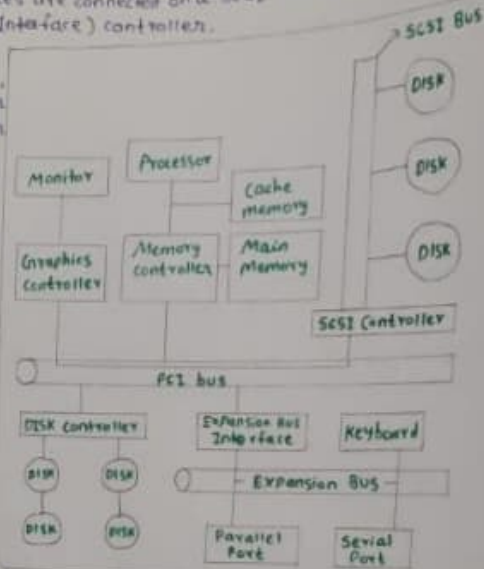
- PCI Slots are utilized to install sound cards, Ethernet and remote cards and presently strong state drives utilizing NVMe innovation to supply SSD drive speeds that are numerous times speedier than SATA SSD speeds. PCI openings too permit discrete design cards to be included to a computer as well.
- PCI openings (and their variations) permit you to include expansions cards to a motherboard. The extension cards increment the machines capabilities past what the motherboard may create alone. Such as: upgraded illustrations, extended sound, expanded USB and difficult drive controller, and extra arrange interface options. to title a couple of.

PCI BUS ARCHITECTURE:

- ## PCI BUS ARCHITECTURE:
- It stands for Peripheral Component Interconnect. PCI Bus Architecture is based on ISA (Industry Standard Architecture) Bus. PCI is a local computer bus for attaching hardware devices to a computer. The PCI bus connects the processor subsystem to the fast devices and an expansion bus that connects relatively slow devices such as a keyboard and a serial and parallel ports. In the upper right position of the figure, three devices are connected on a SCSI bus that is plugged into a SCSI (Small Computer System Interface) controller.
- SCSI BUS

- A controller is a collection of electronics that are operate a port, a bus or a device. A serial port controller is an example of single device controller. It is a single chip in a computer that controls the signals on the wire of a serial ports. The controller has one or more registers for data and control signals.

The processor communicates with the controller by reading and writing bit patterns in those registers.



Advantages of PCI:

- You'll interface a greatest of five components to the PCJ and you'll be able moreover support each of them by settled gadgets on the motherboard.
- The PCJ transport will improve the speed of the exchanges from 33 MHz to 133 MHz with a transfer rate of 1 GBps.

② Disadvantages of PCI:

- 1) PC1 Graphics card cannot get to System memory
- 2) PC1 does not support pipeline.

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RISC PROCESSOR

→ RISC stands for Reduced Instruction Set Computer. In Reduced Instruction Set computer (RISC) architecture, the instruction set of the computer is simplified to reduce the execution time. RISC has a small set of instructions, which generally include register-to-register operations.

→ The data is stored in processor registers for computations, and results of the computations are transferred to the memory using store instructions. All operations are performed within the registers of the CPU. In RISC, all instructions have simple register addressing and hence use less number of addressing modes.

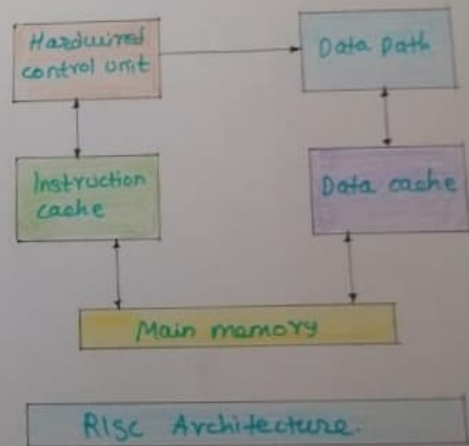
→ RISC uses relatively a simple instructions format and is easy to decode. Here, the instruction length can be fixed and aligned on word boundaries. The RISC processors can execute one instruction per clock cycle. RISC processor is implemented using the hardwired control unit. The hardwired control unit produces control signals which regulate the working of processor's hardware. RISC architecture emphasizes on using the registers rather than memory.

→ This is because the registers are the 'fastest' available memory source. The registers are physically small and are placed on the same chip where the ALU and the control unit are placed on the processor. The RISC instructions operate on the operands present in processor's registers.

→ It is also because all instructions in RISC are simple and execute one instruction per cycle. So, here the instructions are hardwired and there is no need for control store. For each operations, we will have as defined hardware. Making an instruction hardwired is making a function or operation in instruction permanent using connected circuits.

① RISC Instruction Set

RISC instructions are simple and are of fixed size. Each RISC instruction engages a single memory word. RISC instructions operate on processor registers only. The instructions that have arithmetic and logic operation should have their operand either in the processor register or should be given directly in the instruction.



DMA (Direct Memory Access)

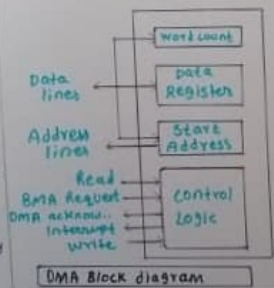
- **Direct Memory Access (DMA)** transfers the block of data between the memory and peripheral devices of the system, without the participation of the processor. The unit that controls the activity of accessing memory directly is called a DMA controller. The processor relinquishes the system bus for a few clock cycles. So, the DMA controller can accomplish the task of data transfer via the system bus.
- **Direct memory access (DMA)** is a mode of data transfer between the memory and I/O devices. This happens without the involvement of the processor. We have studied two other methods of data transfer: Programmed I/O and Interrupt driven I/O.
- In **Programmed I/O**, the processor keeps on scanning whether any device is ready for data transfer. If an I/O device is ready, the processor fully dedicates itself in transferring the data between I/O and memory. It transfers data at a high rate, but it can't get involved in any other activity during data transfer. This is the major drawback of programmed I/O.
- In **Interrupt driven I/O**, whenever the device is ready for data transfer, then it raises an interrupt to processor. Processor completes executing its ongoing instruction and saves its current state. It then switches to data transfer which causes a delay. Here, the processor doesn't keep scanning for peripheral ready for data transfer. But, it is fully involved in the data transfer process. So, it is also not an effective way of data transfer.

* DMA controller transfers the data in three modes.

1. **Burst mode:-** Here, once the DMA controller gains the charge of the system bus, then it releases the system bus only after completion of data transfer. Till then the CPU has to wait for the system buses.
2. **Cycle stealing mode:-** In this mode, the DMA controller forces the CPU to stop its operation and relinquish the control over the bus for a short term to DMA controller.
3. **Transparent mode:-** Here, the DMA controller takes the charge of system bus only if the processor does not require the system bus.

→ DMA controller is a hardware unit that allows i/o devices to access memory directly without the participation of the

1. whenever an I/O device wants to transfer the data to or from memory it sends the DMA requests (DRA) to the DMA controller. DMA controller accepts this DRA and asks the CPU to hold for a few clock cycles by sending it the Hold request (HLB).
2. CPU receives the Hold request (HLB) from DMA controller and relinquishes the bus and sends the Hold acknowledgement (HLDA) to DMA controller.
3. After receiving the Hold acknowledgement (HLDA), DMA controller acknowledges I/O device (convey) that the data transfer can be performed and DMA controller takes the charge of the system bus and transfers the data to or from memory.
4. When the data transfer is accomplished, the DMA raises an interrupt to let know the processor that the task of data transfer is finished & the processor can take control over the bus again and start processing where it has left.



1. The first information is whether the data has to be read from memory or data has to be written to the memory. It passes this info via read or write control lines that is between the processor and DMA controllers control logic.
2. The processor also provides the starting address of the data block in the memory, from where the data block in memory has to be read or where the data block has to be written in memory. DMA controller stores this in its address register. It is also called the Starting address register.
3. The processor also sends the word count, i.e. how many words are to be read or written. It stores this information in the data count or the word count register.
4. The most important is the address of I/O device that wants to read or write data. This information is stored in the data register.

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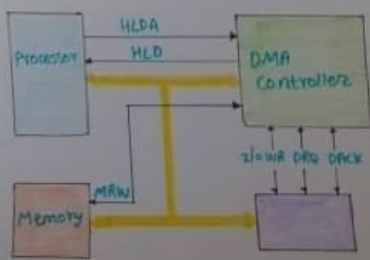
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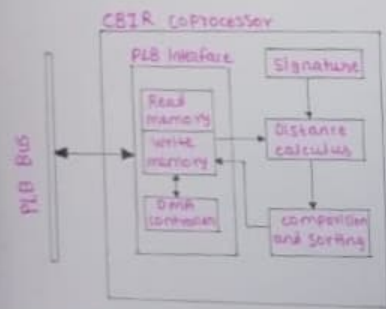


DMA Controller Data Transfer

Coprocessor

It is microprocessor chip. new circuitry can be added with special purpose to perform special tasks or to perform operations on numbers in orders to offload the work of the core CPU. The CPU can then work faster.

We may use a conveyor belt to do some extra work while motor is running. So, the motor is more effectively utilized. Similarly, supplementary processor, i.e., a coprocessor handles the mathematical part of the work when we run complex applications.



* Properties of coprocessors:

1. Without primary microprocessor, coprocessor cannot function.
2. Main processor has to identify and segregate computationally intensive instructions in a program.
3. The instructions which have an intensive amount of calculations are performed by coprocessor.
4. The main processor handles...

① A Co-Processor is many times referred as a Math Processor. As the coprocessor performs routine mathematical tasks, the core processor is freed up from this computation and its time is saved. By taking specialized processing tasks from core CPU, coprocessor reduces the strain on the main microprocessor so that it can run at a greater speed.

② A coprocessor can perform special tasks like complex mathematical calculation or graphical display processing. They perform such jobs faster than core CPU. As a result, overall computer speed of the system increases.

③ To an ARM processor, we can attach the coprocessors. A coprocessor when added, we need to expand instruction set of core CPU or add configurable registers to increase the processing power. The coprocessor interface permits a couple of coprocessor to be connected to the ARM CPU.

* How does coprocessor work?

Coprocessor assists the system in running more efficiently by offloading specific tasks of the CPU. They can be:

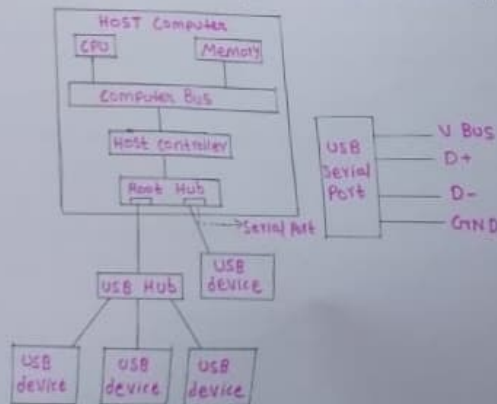
1. Independent type: It works in an asynchronous manner with the CPU. Whenever it performs any task (some calculations, that is not synchronized with the CPU, co-processor can take decisions independently here.

→ CPU can then do its own work without worrying about co-processor's work. Instead of synchronously waiting, CPU that issued that request is freed up to perform another task.

2. Direct control type: Direct control type such as floating-point units - controlled by coprocessor's instructions that are CPU instruction set's part. The CPU that initiated the request waits or checks until co-processor completes operation. The coprocessor is not independent here & is governed by the main CPU. CPU & coprocessor works in synchronization here.

USB (Universal Serial Bus).

- A Universal Serial Bus (USB) is a common interface that enables communication between devices and a host controller such as a personal computer (PC) or smartphone. It connects peripheral devices such as digital cameras, mice, keyboards, printers, scanners, media devices, external hard drives and flash drives. Because of its wide variety of uses, including support for electric power, the USB has replaced a wide range of interfaces like the parallel & serial port.
- A USB is intended to enhance plug-and-play and allow hot swapping. Plug-and-play enables the operating system (OS) to spontaneously configure and discover a new peripheral device without having to restart the computer.
- There are several types of USB connectors. In the past the majority of USB cables were one of two types, type A and type B. The USB 2.0 standard is type A; it has a flat rectangle interface that inserts into a hub or USB host which transmits data and supplies power. A keyboard or mouse are common examples of a type A USB connectors. A type B USB connector is square with beveled exterior corners. It is connected to an upstream port that uses a removable cable such as a printer. The type B connector also transmits data and supplies power. Some type B connectors do not have a data connection and are used only as a power connection.



* USB Architecture:

The USB System is logically a tree topology but physical is a star topology because each USB device communicates directly with the Root Hub. There is only one host controller in any USB system. A USB system consists of USB host controller, Root Hub, Hub, USB cable, USB device, client software and host controller software.

* Host controller:

The host controller initiates all data transfer, and the root hub provides a connection between devices and the host controller. The root hub receives transactions generated by the host controller and transmits them to the USB devices. The host controller uses polling to detect a new device that connected to the bus or is disconnected from it.

* Root Hub:

The root hub performs power distribution to the devices, enables and disables the ports, and reports the status of each port to the host controller. The root hub provides the connection between the host controller and USB ports.

* Hub:

Hubs are used to expand the number of devices connected to the USB system. Hubs can detect when a device is attached or removed from the port.

* USB cable:

A USB port has four pins, which consists of four wires. The V bus is used to power the devices.

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Synchronous Bus:-

In this Synchronous Bus operations are synchronized with reference to a clock signal. The bus clock is generally derived from the Computer System Clock, however System with a processor clock of over 500MHz.

Buses were traditionally slower than processors because memory access times are typically longer than processor clock cycles. A bus transaction often takes several clock cycles, although the cycles are collectively referred to by many as a bus cycle.

As Read operation's on a Synchronous Bus are as given in form of a Diagram here.

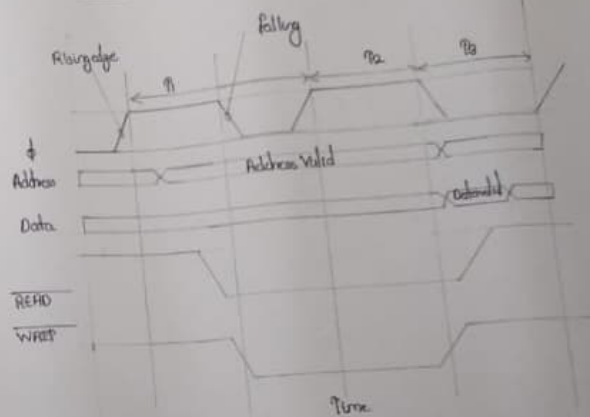


Fig 1:- Read operation on a Synchronous Bus

Asynchronous Bus:-

This Bus has no System clock. Handshaking is done to properly conduct the transmission of data between the sender and the receiver.

The process is illustrated in a given Figure (Diagram) provided

Note that there is no clock and that starting and ending of the data transfer are indicated by Special Synchronization signals.

As Synchronous buses are typically faster than asynchronous buses because there is no overhead to establish a time reference for each transaction.

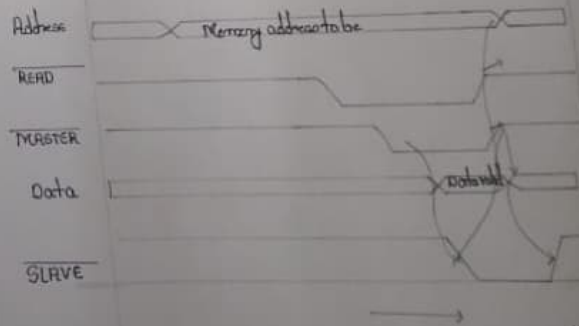


Fig 2:- Read Operation on a Asynchronous Bus

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SUPER SCALAR PROCESSOR...

A more aggressive approach is to equip the processor with multiple processing units to handle several instructions in parallel in each processing cycle. With this arrangement, several instructions start execution in the same clock cycle and the process is said to use multiple issue. Such processors are capable of achieving an instruction execution throughput of more than one instruction per cycle. They are known as 'Superscalar processors'.

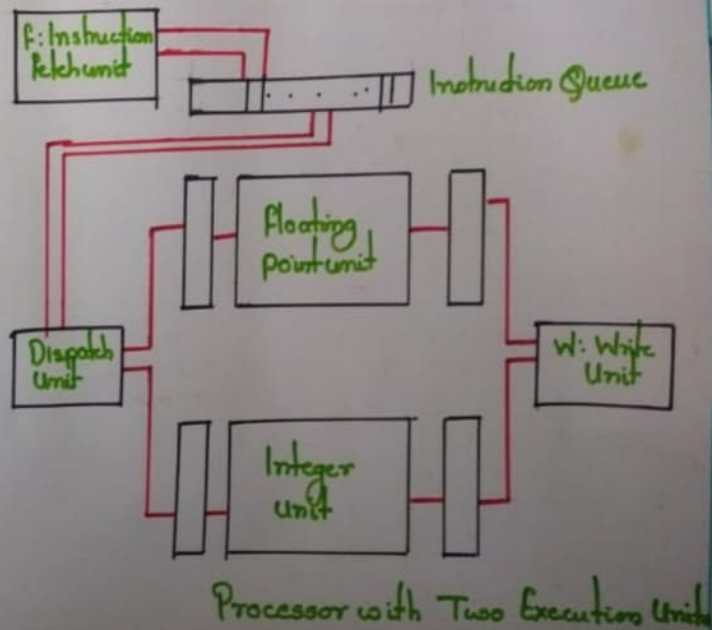
In the above diagram, there are a processor with two execution units, one for integers and one for floating point operations. The instruction fetch unit is capable of reading the instructions at a time and storing them in the instruction queue.

Advantages:-

1. Higher Instruction Throughput
2. Better Utilization of CPU resources
3. Instruction-level Parallelism
4. Reduced program execution time
5. No need for special programming.

Disadvantages:-

1. Complex hardware design
2. Higher power consumption
3. Diminishing Returns
4. Large silicon area cost
5. Challenging hazard handling.



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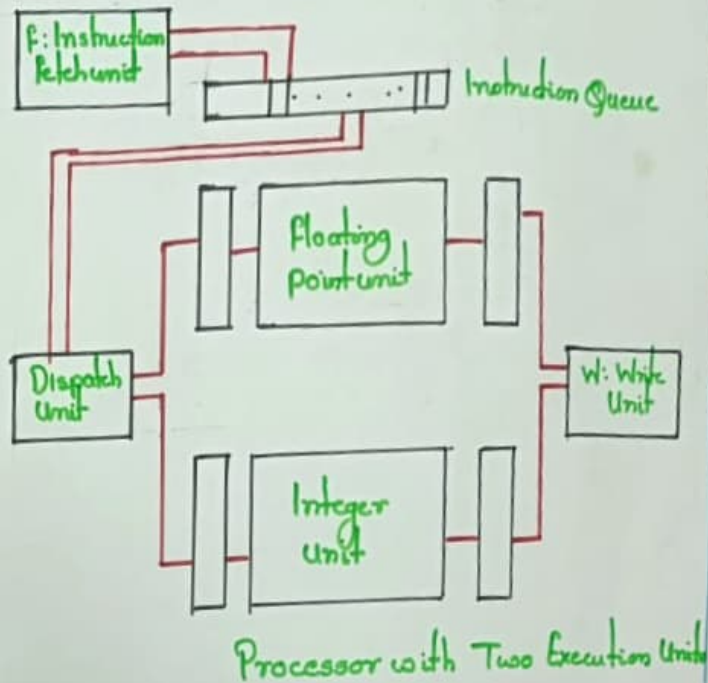
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Processor with Two Execution Units

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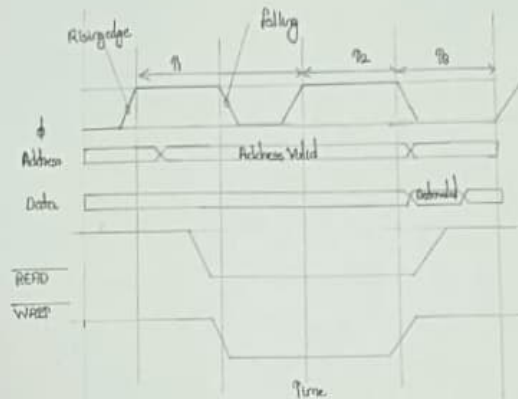


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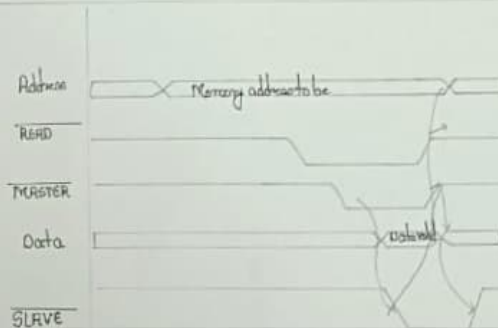


Fig 2:- Read Operations on a Asynchronous Bus.

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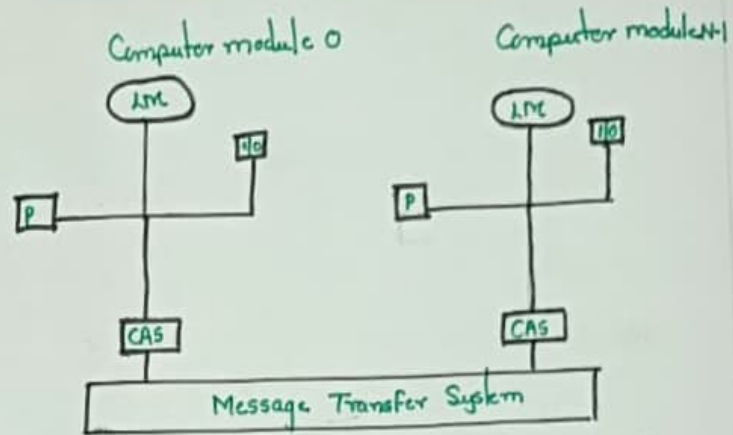
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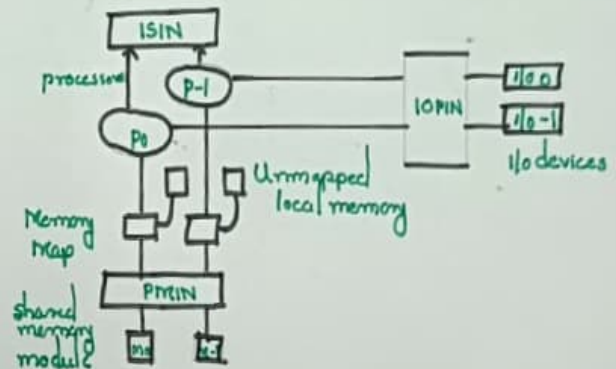
Loosely Coupled Multiprocessor Architecture.

1. Every CPU has its memory module in a loosely Coupled System.
2. Memory Conflicts is rare in this type of Systems.
3. It is very efficient when the processor executes on several CPUs have minimal interaction between them.
4. Its interconnection network is the Message Transfer System (MTS).
5. It is less costly but bigger in size.
6. Its data rate is low.
7. The power consumption of the coupled system is high.



Tightly Coupled Multiprocessor Archi-(System)

1. CPU shares the memory module in a tightly coupled System.
2. Memory conflict is common in these types of Systems.
3. Its interconnections are the (PPEIN), (IOPIN) and (ISIN).
4. It is more costly but smaller in size.
5. Its data rate is high.
6. The power consumption of the tightly coupled system is low.



Programmed I/O

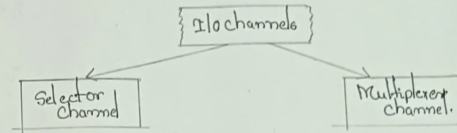
1. In programmed I/O device in sequence and in effect ask each one if it needs communication with the processor. This checking is achieved by continuous polling cycles and hence processor cannot execute another cycle in sequence.
2. It is implemented without interrupt hardware support.
3. It does not depend on interrupt status.
4. It does not need initialization of stack.
5. System throughput decreases as number of I/O devices connected in the system increases.

Interrupt Driven I/O

1. The external asynchronous input is used to tell the processor that I/O device needs its service and hence processor does not have to check whether I/O devices needs its service or not.
2. It is implemented using interrupt hardware support.
3. Interrupt input is enabled to process interrupt driven I/O.
4. It needs initialization of stack.
5. System throughput does not depend on the system.

I/O Channels - It is an extension of the DMA concept. It has ability to execute I/O instructions using special purpose processor on I/O channel and complete control over I/O operations.
Program Specifier - Device or devices, Area or areas of memory, priority, and error condition actions.

Types of I/O channels



1. **Selector Channel**:- Selector Channel controls multiple high speed devices. It is dedicated to the transfer of data with one of the devices. In selector channel, each device is handled by a Controller or I/O module. It controls the I/O controllers shown in the figure.
2. **Multiplexer Channel**:- Multiplexer channel is a DMA Controller that can handle multiple devices at the same time. It can do block transfers for several devices at once. There are two types of multiplexers are used in this channel:-

1. **Byte Multiplexer**:-
 It is used for low-speed devices. It transmits or accepts characters.

2. **Block Multiplexer**:-
 It accepts or transmits blocks of characters. It interleaves blocks of bytes from several devices.

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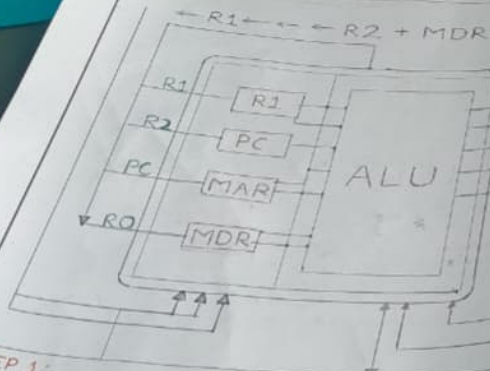
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DATA TRANSFER
MICRO-OPERATION

ARITHMETIC
MICRO-OPERATION

LOGIC
MICRO-OPERATION

FETCH-DECODE-EXECUTE CYCLE



STEP 1:

Fetch-PC → MAR
Memory → MDR

STEP 2:

Decode-MDR →
Instruction Register

STEP 3:

$R1 \leftarrow R2 + R3$
 $R1 \leftarrow R2 \text{ AND } R3$
MAR ↔ PC

CONTROL
UNIT

DELTA

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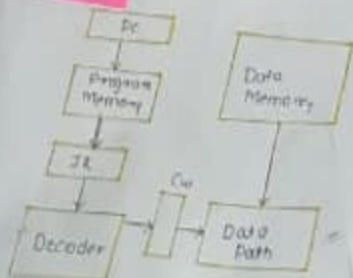
RISC

(Reduced Instruction Set Computer)

Simple
often use
emergence



Feature
simplify
emergence



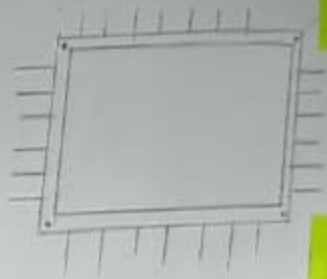
Real world
example
ARM

- Advantages -
- Faster execution
 - Efficiency & cost
 - Design

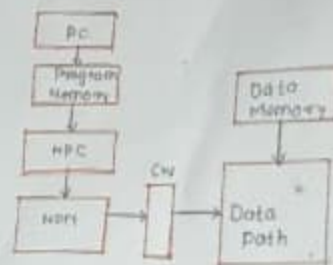
CISC

(Complex Instruction Set Computer)

Complex
often use
emergence



Feature
simplify
emergence



Real world
example -
Intel x86

- Advantages -
- Reduced code size
 - Easier for compilers
 - Versatile instructions

Notes -
Extended words
Support Java
Memory fast
Execution fast